## Principal Component Analysis

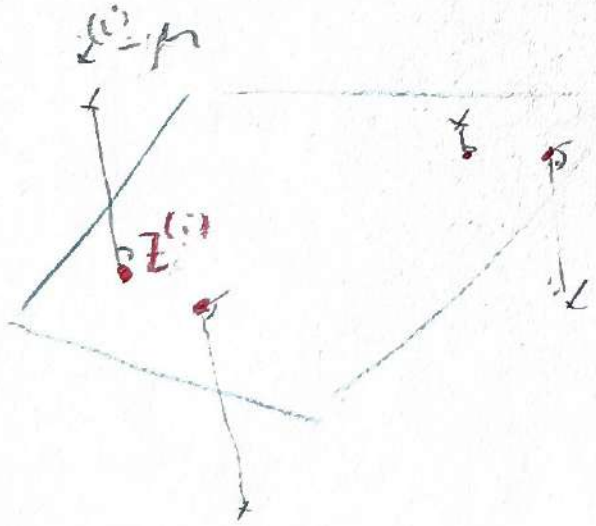
Idea: reduce dimension of the data
- save memory, training time
- reduce overfitting

PCA:

- linear: eigenvectors of the covariance matrix
- nonlinear: autoencoders

Original data: $D$ dimensional $\quad$ mean $\mu = \frac{1}{N}\sum_i x^{(i)}$

Projection to $K$ dim. $S$ subspace




$\underline{U}$ : orthonormal base of $S$

$\underline{z}^{(i)} = \underline{U}^T (\underline{x}^{(i)} - \mu)$

$\underline{z}^{(i)}$ representation of $\underline{x}^{(i)}$

Reconstruction $\tilde{x}^{(i)} = \mu + \underline{U}\,\underline{z}^{(i)}$

Reconstruction error: $\frac{1}{N}\sum_{i=1}^{N} \| \underline{x}^{(i)} - \tilde{\underline{x}}^{(i)} \|^2$ to be minimized

The subspace $S$ is useful if the data is well scattered. This gives us the chance that the data will be separable

Second criteria: $\max \sum_{j=1}^{k} \text{var}(z_j)$ variance in each direction

$$\max \sum_{j=1}^{k} \text{Var}(z_j) = \frac{1}{N} \sum_{j=1}^{k} \sum_{i=1}^{N} (z_j^{(i)} - \bar{z}_j)^2 = \frac{1}{N} \sum_i \|\underline{z}^{(i)} - \underline{\bar{z}}\|^2 =$$

$$= \frac{1}{N} \sum_i \|\underline{z}^{(i)}\|^2, \text{ since } \underline{\bar{z}} = 0$$

↑ we removed the mean $\underline{x}$

Let us show that the two criteria are equivalent. We will show that the sum of them is constant which is enough:

$$\text{We use } \underline{\Sigma} = \|\underline{x}^{(i)} - \underline{\mu}\| = \|\underline{U}\underline{z}^{(i)} + \underline{\mu} - \underline{\mu}\| = \|\underline{U}\underline{z}^{(i)}\| = \|\underline{z}^{(i)}\|$$

$$\frac{1}{N} \sum_{i=1}^{N} \|\underline{x}^{(i)} - \underline{\tilde{x}}^{(i)}\|^2 + \frac{1}{N} \sum_{i=1}^{N} \|\underline{z}^{(i)}\|^2 = \frac{1}{N} \sum_{i=1}^{N} \|\underline{x}^{(i)} - \underline{\tilde{x}}^{(i)}\|^2 +$$

$$+ \frac{1}{N} \sum_{i=1}^{N} \|\underline{\tilde{x}}^{(i)} - \underline{\mu}\|^2$$

First term: is perpendicular to $S$

Second term: is inside $S$

$$S = \frac{1}{N} \sum_{i=1}^{N} \|\underline{x}^{(i)} - \underline{\tilde{x}}^{(i)} + \underline{\tilde{x}}^{(i)} - \underline{\mu}\|^2 = \frac{1}{N} \sum_{i=1}^{N} \|\underline{x}^{(i)} - \underline{\mu}\|^2$$

const. independent of $S$ ↵

PCA: maximize variance

Spectral decomposition:

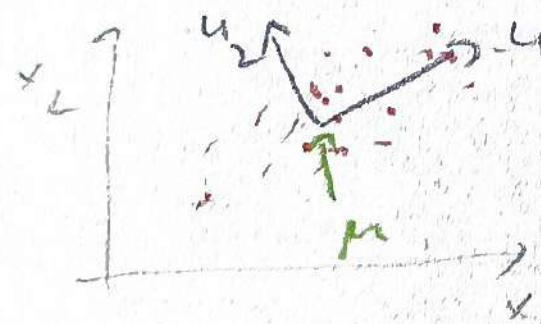
$\underline{\underline{A}} = \underline{\underline{Q}}\,\underline{\underline{\Lambda}}\,\underline{\underline{Q}}^T$ $\underline{\underline{Q}}$: columns are eigenvectors

$\underline{\underline{\Lambda}}$: diagonals are eigenvalues

$\lambda_i \in \mathbb{R} \iff \underline{\underline{A}}$ symmetric

Covariance matrix: "dyadic product"

$$\Sigma = \frac{1}{N}\sum_{i=1}^{N}(\underline{x}^{(i)} - \underline{\mu})(\underline{x}^{(i)} - \underline{\mu})^T = \frac{1}{N}\sum_{i=1}^{N}(\underline{x}^{(i)} - \underline{\mu}) \circ (\underline{x}^{(i)} - \underline{\mu})$$




It is almost like a moment of inertia. The magnitude of the eigenvalue indicates the importance of that direction

Eigenvectors are called principle components

For $K=1$ we can prove that the largest eigenvalue and its eigenvector is the solution

$$\frac{1}{N}\sum_i \|z^{(i)}\|^2 = \max = \frac{1}{N}\sum_i \left(\underline{u}^T(\underline{x}^{(i)} - \underline{\mu})\right)^2 =$$

$$= \frac{1}{N}\sum_i \underline{u}^T(\underline{x}^{(i)} - \underline{\mu})(\underline{x}^{(i)} - \underline{\mu})\,\underline{u} = \underline{u}^T\left(\frac{1}{N}\sum_i (\underline{x}^{(i)} - \underline{\mu})(\underline{x}^{(i)} - \underline{\mu})\right)\underline{u} =$$

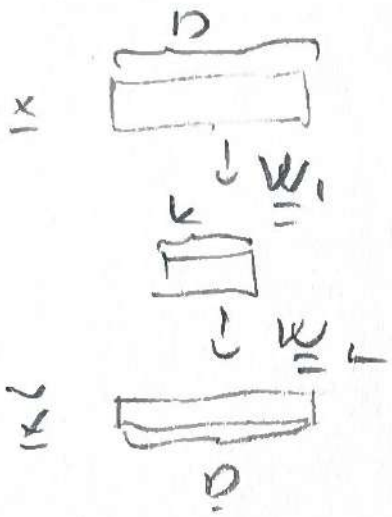
$$= \underline{u}^T\underline{\underline{\Sigma}}\,\underline{u} = \underline{u}^T\,\underline{\underline{Q}}\,\underline{\underline{\Lambda}}\,\underline{\underline{Q}}^T\underline{u} = \underline{a}^T\underline{\underline{\Lambda}}\,\underline{a} = \sum_j \lambda_j a_j^2 \qquad \left(\underline{a} = \underline{\underline{Q}}^T\underline{u}\right)$$

max if $\underline{u}$ is the eigenvector of the largest eigenvalue

# Autoencoders

- Linear autoencoders: $D \times K$ matrix does the trick: $\underline{x}\,\underline{\underline{W}} = \underline{z}$ $\quad \underline{\underline{W}}$ : $D \times K$ matrix

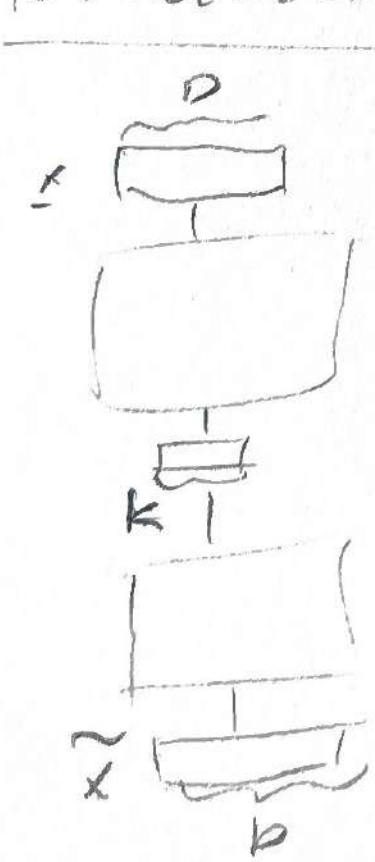
Use neural network to determine $\underline{\underline{W}}$




$$\underline{\tilde{x}} = \underline{\underline{W}}_2\,\underline{\underline{W}}_1\,\underline{x}$$

$$\mathcal{L} = \|\underline{x} - \underline{\tilde{x}}\|^2$$

- Nonlinear autoencoders:







Teaching: input and output are the same

Autoencoder: unsupervised learning
- text processing
- noise reduction
- keyword detection
- clustering